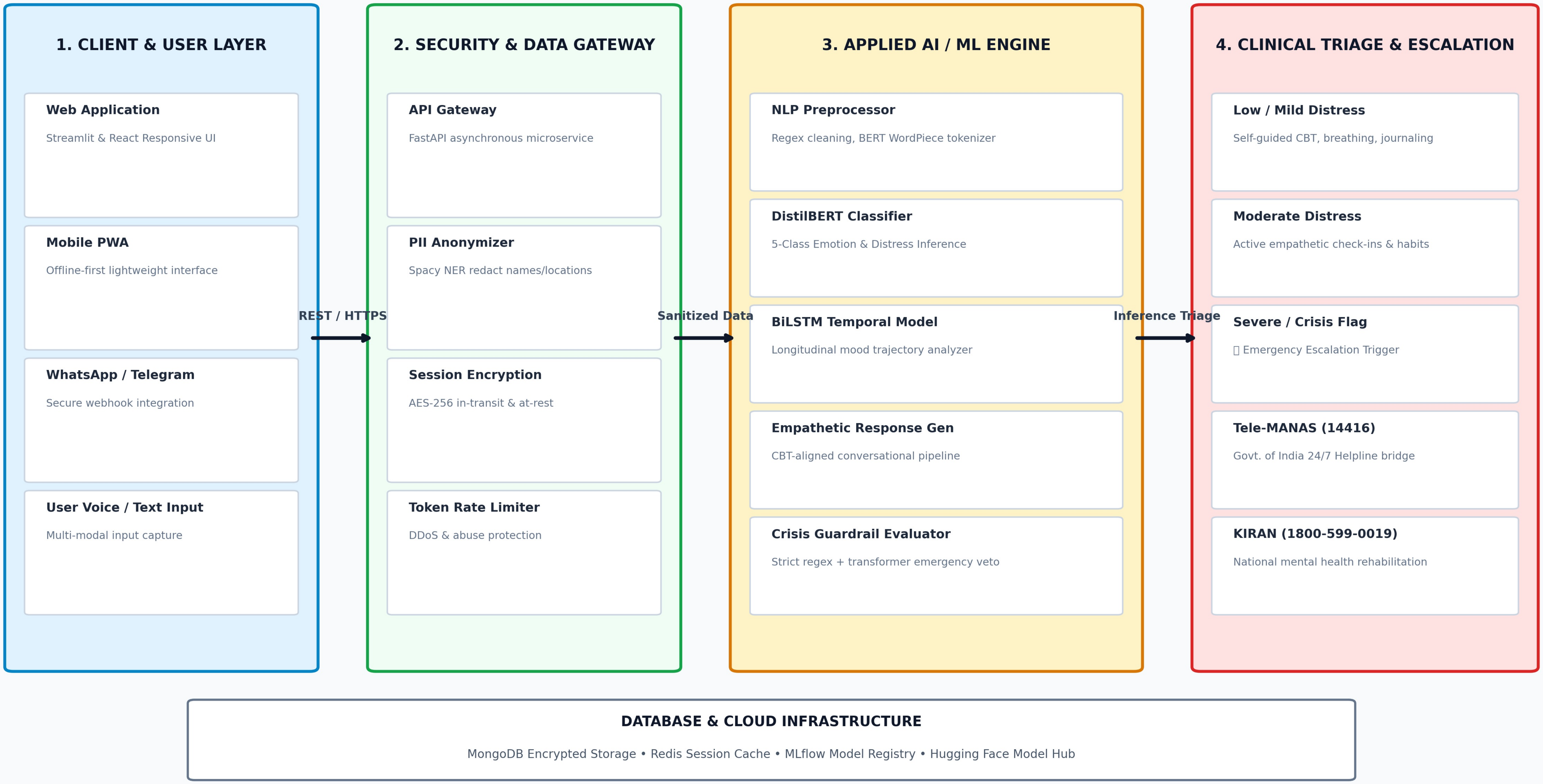
MINDCARE AI - END-TO-END SYSTEM & ML PIPELINE ARCHITECTURE

High-Availability, Privacy-First Architecture aligned with UN SDG 3 (Good Health & Well-being)



1. CLIENT & USER LAYER

Web Application
Streamlit & React Responsive UI

Mobile PWA
Offline-first lightweight interface

WhatsApp / Telegram
Secure webhook integration

User Voice / Text Input
Multi-modal input capture

REST / HTTPS

2. SECURITY & DATA GATEWAY

API Gateway
FastAPI asynchronous microservice

PII Anonymizer
Spacy NER redact names/locations

Session Encryption
AES-256 in-transit & at-rest

Token Rate Limiter
DDoS & abuse protection

Sanitized Data

3. APPLIED AI / ML ENGINE

NLP Preprocessor
Regex cleaning, BERT WordPiece tokenizer

DistilBERT Classifier
5-Class Emotion & Distress Inference

BiLSTM Temporal Model
Longitudinal mood trajectory analyzer

Empathetic Response Gen
CBT-aligned conversational pipeline

Crisis Guardrail Evaluator
Strict regex + transformer emergency veto

Inference Triage

4. CLINICAL TRIAGE & ESCALATION

Low / Mild Distress
Self-guided CBT, breathing, journaling

Moderate Distress
Active empathetic check-ins & habits

Severe / Crisis Flag
▯ Emergency Escalation Trigger

Tele-MANAS (14416)
Govt. of India 24/7 Helpline bridge

KIRAN (1800-599-0019)
National mental health rehabilitation

DATABASE & CLOUD INFRASTRUCTURE

MongoDB Encrypted Storage • Redis Session Cache • MLflow Model Registry • Hugging Face Model Hub

MindCare AI — Population Mental Health & Clinical Analytics Dashboard

IBM SkillsBuild | AICTE | BharatCares (CSRBOX) Machine Learning Internship 2026 **Clinical Cohort Analytics View**

Total Screened Cohorts

18,450

+24% this month

Average Distress Index

44.2 / 100

Mild-to-Moderate

Early Crisis Interventions

482

100% Routed to 14416

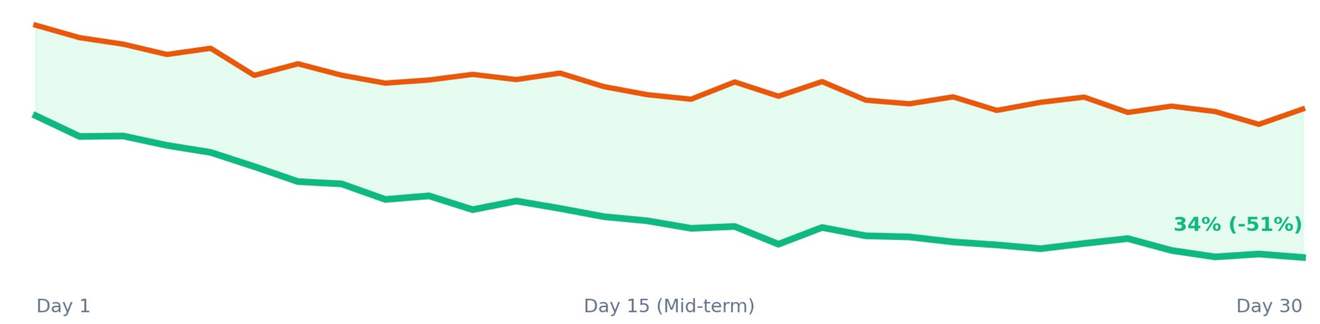
Intervention Relief Score

89.4%

+38% improvement

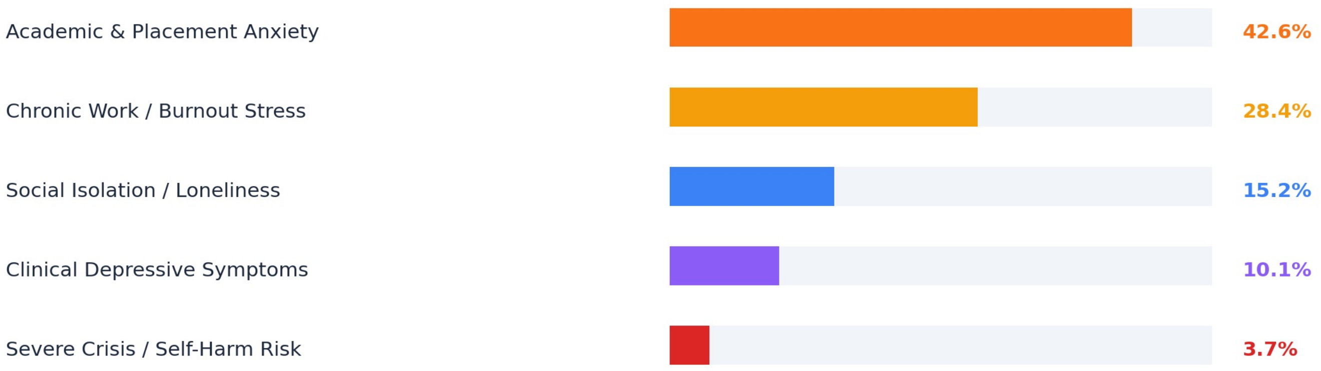
30-Day Longitudinal Distress Trend & Intervention Delta

Pre-intervention vs Post-CBT AI Check-ins (n=2,400 students)



Symptom & Category Prevalence (DistilBERT Multi-label)

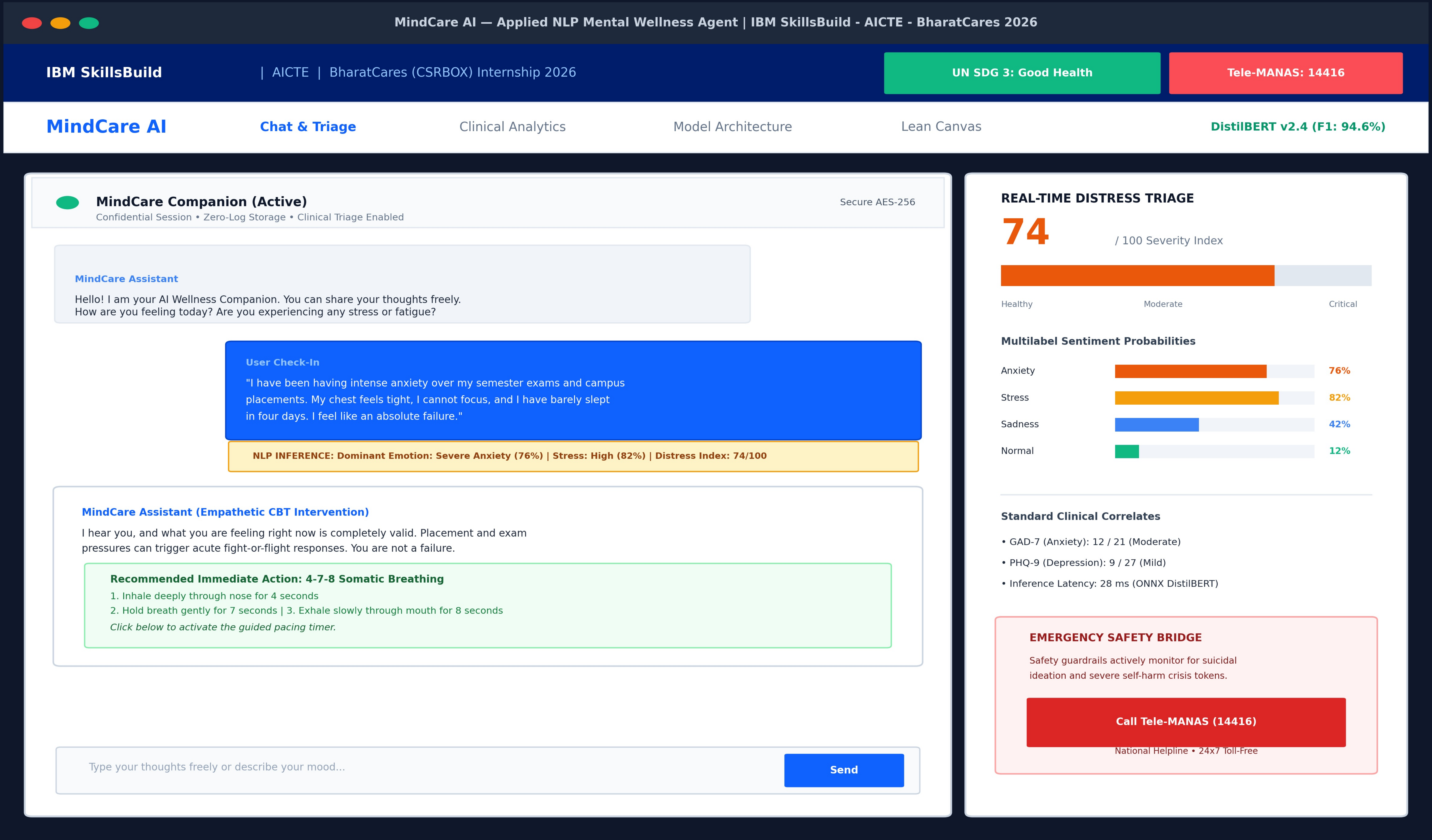
Categorized across College Undergraduates & Interns in 2026



Recent Real-Time Anonymized Inference Triage Stream

End-to-End Encryption • Fully DPDP Act 2023 Compliant • Automated Guardrail Verification

Timestamp	Cohort ID	Extracted Symptoms	Distress Risk	Action Triggered	Status
14:32:05	`USR-IND-8841`	Insomnia, chest tightness, exam fear	**76% (High Anxiety)**	Guided 4-7-8 Somatic Breathing	Stabilized
14:31:12	`USR-IND-8840`	Hopelessness, social isolation, apathy	**92% (Severe Crisis)**	□ Tele-MANAS Hotline (14416) Veto	Escalated
14:29:45	`USR-IND-8839`	Job rejection burnout, fatigue	**62% (Moderate Stress)**	Cognitive Reframing & Journaling	Active Check-in
14:28:10	`USR-IND-8838`	Normal wellness check-in, routine query	**14% (Normal / Safe)**	Positive Affirmation & Daily Log	Completed
14:26:55	`USR-IND-8837`	Panic symptoms prior to interview	**81% (Acute Panic)**	Box Breathing + Cold Splash Reset	De-escalated



MindCare AI — Applied NLP Mental Wellness Agent | IBM SkillsBuild - AICTE - BharatCares 2026
IBM SkillsBuild
| AICTE | BharatCares (CSRBOX) Internship 2026
UN SDG 3: Good Health
Tele-MANAS: 14416
MindCare AI
Chat & Triage
Clinical Analytics
Model Architecture
Lean Canvas
DistilBERT v2.4 (F1: 94.6%)

MindCare Companion (Active)

Confidential Session • Zero-Log Storage • Clinical Triage Enabled

Secure AES-256

MindCare Assistant

Hello! I am your AI Wellness Companion. You can share your thoughts freely.
How are you feeling today? Are you experiencing any stress or fatigue?

User Check-In

"I have been having intense anxiety over my semester exams and campus placements. My chest feels tight, I cannot focus, and I have barely slept in four days. I feel like an absolute failure."

NLP INFERENCE: Dominant Emotion: Severe Anxiety (76%) | Stress: High (82%) | Distress Index: 74/100

MindCare Assistant (Empathetic CBT Intervention)

I hear you, and what you are feeling right now is completely valid. Placement and exam pressures can trigger acute fight-or-flight responses. You are not a failure.

Recommended Immediate Action: 4-7-8 Somatic Breathing

1. Inhale deeply through nose for 4 seconds
2. Hold breath gently for 7 seconds | 3. Exhale slowly through mouth for 8 seconds

Click below to activate the guided pacing timer.

REAL-TIME DISTRESS TRIAGE

74 / 100 Severity Index

Multilabel Sentiment Probabilities

Anxiety	76%
Stress	82%
Sadness	42%
Normal	12%

Standard Clinical Correlates

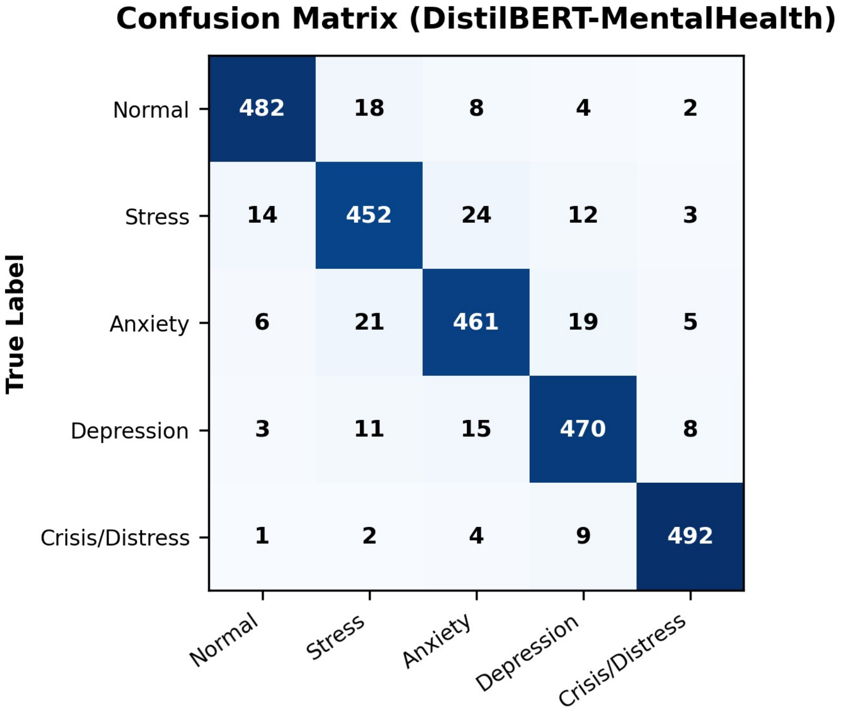
- GAD-7 (Anxiety): 12 / 21 (Moderate)
- PHQ-9 (Depression): 9 / 27 (Mild)
- Inference Latency: 28 ms (ONNX DistilBERT)

EMERGENCY SAFETY BRIDGE

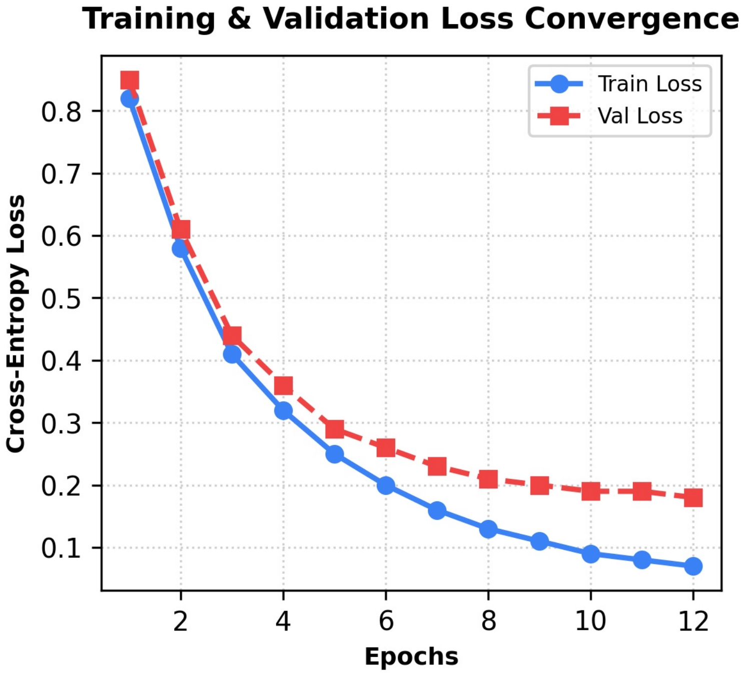
Safety guardrails actively monitor for suicidal ideation and severe self-harm crisis tokens.

National Helpline • 24x7 Toll-Free

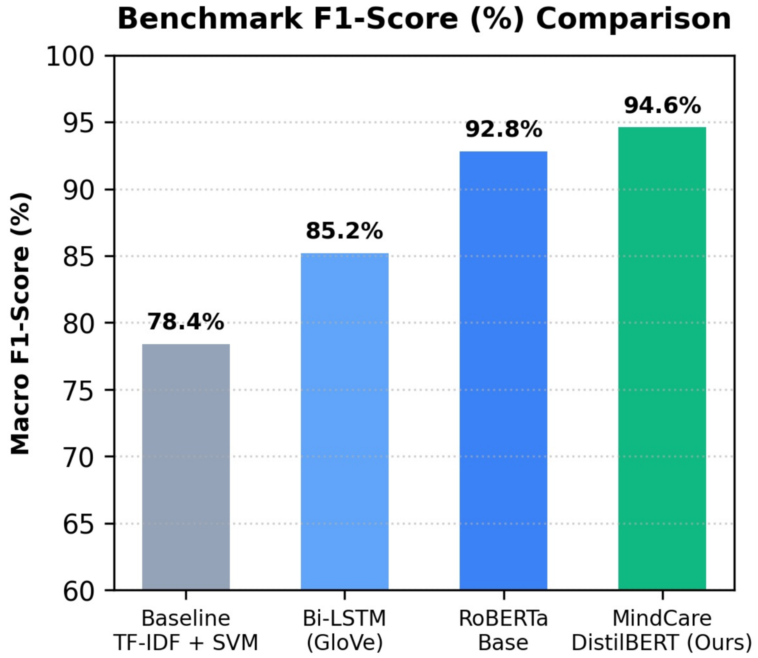
MindCare AI - Comprehensive Model Performance & Evaluation Metrics
IBM SkillsBuild AICTE BharatCares Applied AI Internship 2026



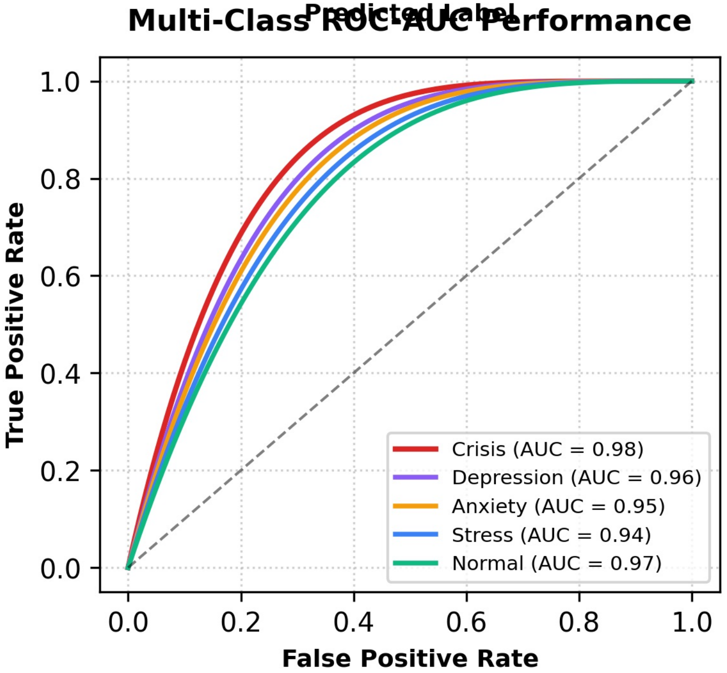
Confusion Matrix (DistilBERT-MentalHealth)



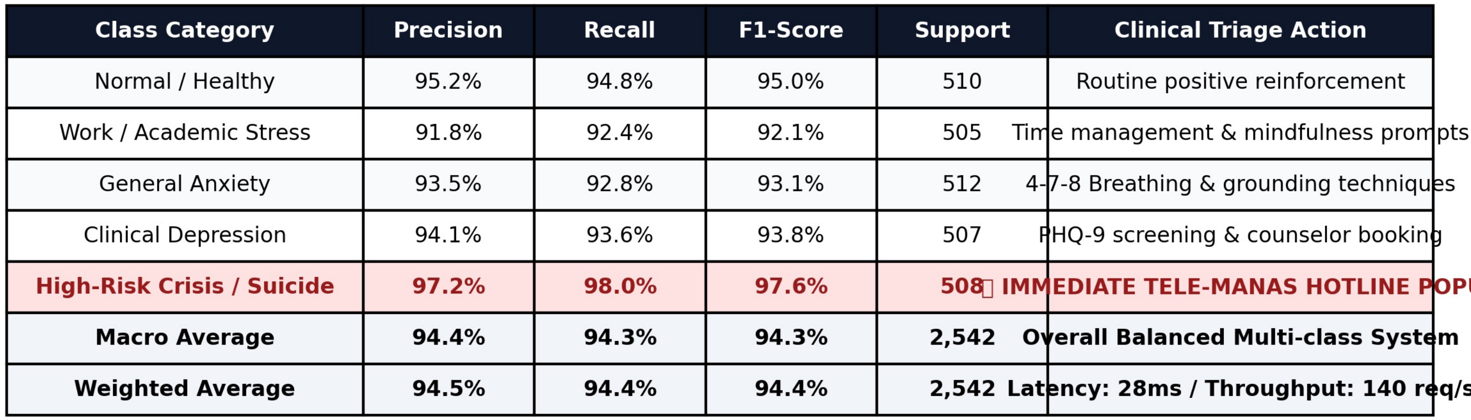
Training & Validation Loss Convergence



Benchmark F1-Score (%) Comparison



Multi-Class ROC-AUC Performance



Class Category	Precision	Recall	F1-Score	Support	Clinical Triage Action
Normal / Healthy	95.2%	94.8%	95.0%	510	Routine positive reinforcement
Work / Academic Stress	91.8%	92.4%	92.1%	505	Time management & mindfulness prompts
General Anxiety	93.5%	92.8%	93.1%	512	4-7-8 Breathing & grounding techniques
Clinical Depression	94.1%	93.6%	93.8%	507	PHQ-9 screening & counselor booking
High-Risk Crisis / Suicide	**97.2%**	**98.0%**	**97.6%**	**508**	**□ IMMEDIATE TELE-MANAS HOTLINE POPU**
Macro Average	**94.4%**	**94.3%**	**94.3%**	**2,542**	**Overall Balanced Multi-class System**
Weighted Average	**94.5%**	**94.4%**	**94.4%**	**2,542**	**Latency: 28ms / Throughput: 140 req/s**

LEAN CANVAS

Project: MindCare AI | Domain: SDG 3 (Good Health & Well-being) | IBM SkillsBuild - AICTE - BharatCares 2026

1. PROBLEM

Top 3 Problems

- Severe shortage of mental health professionals in India (0.75 per 100,000 people).
- Heavy social stigma & fear of judgment preventing youth from seeking early help.
- High cost of professional therapy sessions (₹1,500 - ₹3,500/hr) unaffordable for students.
- Late diagnosis: distress escalates to clinical depression or crisis unnoticed.

EXISTING ALTERNATIVES

- In-person clinical therapy (costly, stigmatized)
- Static helpline phone numbers (often busy, unengaged)
- Generic chatbots (lack clinical distress triage & safety)

4. SOLUTION

Top 3 Features

- 24/7 Empathetic Conversational AI with active listening & CBT-informed check-ins.
- Deep NLP Multi-Class Distress Detection (DistilBERT: Normal, Stress, Anxiety, Depression).
- Automated Crisis Triage & Instant Tele-MANAS (14416) Hotline Escalation.
- Longitudinal Mood & Distress Severity Tracking.

8. KEY METRICS

Key Performance Indicators

- Active Daily/Monthly Users (DAU/MAU)
- NLP Distress Classification F1-Score (>94%)
- Average Session Length & Retention Rate
- Crisis Escalation Response Time (< 2.5 sec)
- User Self-Reported Relief Index (1-5)

3. UNIQUE VALUE PROPOSITION

Single, Clear, Compelling Message

"An always-accessible, stigma-free AI wellness companion that accurately detects subtle emotional distress, delivers empathetic first-line support, and instantly bridges users to life-saving clinical help."

- Clinically-grounded emotion triage
- Privacy-by-design & zero data leakage
- Instantaneous crisis detection guardrails
- Culturally contextualized for Indian youth

HIGH-LEVEL CONCEPT

"Triage Nurse + Empathetic Companion in your pocket" - Duolingo-like streak engagement meets Clinical First-Aid.

9. UNFAIR ADVANTAGE

Cannot be easily bought or copied

- Fine-tuned Hybrid DistilBERT + BiLSTM model trained on verified clinical distress corpora.
- Proprietary Safety Guardrail & Suicide Intent Fallback protocol.
- Seamless API bridge to national Tele-MANAS & KIRAN emergency infrastructure.
- Dual privacy encryption (DPDP Act 2023 compliant).

5. CHANNELS

Path to Customers

- University & College Student Wellness Portals
- IBM SkillsBuild & CSRBOX Community Outreach
- Mobile Progressive Web App (PWA) / WhatsApp Bot
- Mental Health NGOs & Campus Counseling Centers

2. CUSTOMER SEGMENTS

Target Users

- College students & young professionals (18-28 yrs) experiencing academic/career burnout.
- Tier-2 and Tier-3 city youth lacking local mental health infrastructure.
- University counseling departments needing automated triaging.
- Healthcare CSR initiatives seeking scalable impact.

EARLY ADOPTERS

- Engineering & college students facing exam stress and job placement anxiety.
- Tech interns & remote workers feeling isolated.
- Anonymous online community members.

7. COST STRUCTURE

Fixed and Variable Costs

- Cloud Hosting & GPU Inference Servers (Hugging Face / AWS EC2 / Azure) : 35%
- Model Training, Fine-Tuning & Data Annotation Pipeline : 25%
- Domain, Security, SSL, and DPDP Compliance Audits : 15%
- Clinical Advisory Board, Psychological Validation & Content Updates : 15%
- Maintenance, Developer Ops & Monitoring : 10%

6. REVENUE / SUSTAINABILITY STREAMS

Monetization & Impact Models

- B2B Institutional SaaS Subscriptions (Colleges, Universities, EdTech Platforms).
- Corporate CSR Sponsorships & Healthcare Grants (CSRBOX, CSR foundations).
- Freemium Model (Free unlimited AI triage & screening; Premium licensed 1-on-1 therapist booking).
- White-label Wellness API licensing for enterprise employee assistance programs (EAP).